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#!/usr/bin/env python3
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"""
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generate_jsonl_markers.py
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PURPOSE FOR BEGINNER DEVELOPERS:

For an AI agent to search through hundreds of files without crashing phone memory, it cannot open every file. It needs an index file (manifest.jsonl).

This script reads every Markdown file in your vault, grabs its title, first paragraph, calculates a unique SHA256 fingerprint, extracts 15 keyword tokens, and writes a single JSON line for each file into manifest.jsonl.

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"""
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import json
```

```
import hashlib
```

```
from pathlib import Path
```

```
from datetime import datetime
```

```
VAULT_DIR = Path("~/novae-xorpus").expanduser()
```

```
OUTPUT_FILE = VAULT_DIR / "manifest.jsonl"
```

```
def make_marker(file_path: Path) -> dict:
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```
    content = file_path.read_text(encoding="utf-8", errors="ignore")
```

```
    title = file_path.stem
```

```
    summary = ""
```

```
    tokens = set()
```

```
    for line in content.split("\n"):
```

```
        stripped = line.strip()
```

```
        if not stripped or stripped.startswith("---"):
```

```
            continue
```

```
        if stripped.startswith("# ") and title == file_path.stem:
```

```
            title = stripped.lstrip("# ").strip()
```

```
        elif not summary and not stripped.startswith("#"):
```

```
            summary = stripped[:200]
```

```
        if stripped.startswith("##"):
```

```
            for w in stripped.lstrip("# ").lower().split():
```

```
                if len(w) > 3 and w.isalnum():
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```
                    tokens.add(w)
```

```
    h = hashlib.sha256(content.encode("utf-8")).hexdigest()[:16]
```

```
    return {
```

```
        "record_id": f"REC_{h}_{file_path.stem.upper()}[:20]",
```

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"path": str(file_path.relative_to(VAULT_DIR)),
"sha256": h,
"title": title,
"summary": summary or "Technical reference document.",
"tokens": list(tokens)[:15],
"last_modified": datetime.fromtimestamp(file_path.stat().st_mtime).isoformat()
}

```

```

def main():
    markers = []
    for md in VAULT_DIR.rglob("*.md"):
        if ".git" in str(md):
            continue
        try:
            markers.append(make_marker(md))
        except Exception as e:
            print(f"[!] Error on {md.name}: {e}")

    with open(OUTPUT_FILE, "w", encoding="utf-8") as f:
        for m in markers:
            f.write(json.dumps(m, ensure_ascii=False) + "\n")

    print(f"[✓] Generated {len(markers)} JSONL markers in {OUTPUT_FILE}")

if __name__ == "__main__":
    main()

```